

FAST National University of Computer and Emerging Sciences

H-11/4, Islamabad

CS3014 - Applied Human Computer Interaction

D0: Team Charter, Problem-Space Brief and Stakeholder Map

Team Name: MSM Technologies

Section: G

Deliverable: D0

Submitted by: Muhammad Umar — 23I-0782

Name	Roll Number	Role
Muhammad Umar	23I-0782	Team Member
Mirza Yahya Adeel	23I-0768	Team Member
Muhammad Usman Al Haq	23I-0585	Team Member
Shahmeer Atif	23I-0711	Team Member

Part 1: Team Charter

1.1 Team Roster

Name	Roll Number
Muhammad Umar	23I-0782
Mirza Yahya Adeel	23I-0768
Muhammad Usman Al Haq	23I-0585
Shahmeer Atif	23I-0711

1.2 Responsibility Matrix

Each stage has a designated coordination lead. The lead is responsible for ensuring the stage's work gets done and is properly evidenced — not for doing it alone.

Project Stage	Coordination Lead	Roll Number
Empathize	Mirza Yahya Adeel	23I-0768
Define	Muhammad Usman Al Haq	23I-0585
Ideate	Mirza Yahya Adeel	23I-0768
Prototype	Muhammad Umar	23I-0782
Test	Shahmeer Atif	23I-0711

1.3 Meeting Cadence

- The team will hold a minimum of one weekly check-in meeting to review progress, assign tasks, and address blockers.
- Additional meetings will be scheduled as deliverable deadlines approach or when a stage requires collaborative decision-making.
- Meetings will be held in person on campus where possible, with online alternatives via a video call when needed.
- Decision-making: decisions will first be discussed openly. Where the team disagrees, a majority vote applies. If the vote is tied, the coordination lead for the current stage has the deciding vote.

1.4 Contribution Tracking Method

- The team will maintain a shared Google Doc contribution log, updated after every meeting and every major task completion.
- Each entry in the log will record: the team member's name, the task completed, the date, and the approximate time spent.
- Document edit history on shared Google Docs will serve as a secondary record of individual contributions.
- This log will be maintained continuously from Week 1 and will be referenced during peer assessment and any individual viva.

1.5 Communication Channel and Norms

- Primary channel: WhatsApp group chat (all four members included).
- Expected response time: all members are expected to respond within 24 hours of a message being sent.
- Escalation process: if a member is unresponsive for 48 hours or more without prior notice, the team lead for the current stage will attempt direct contact (phone call). If the member remains unresponsive after a further 24 hours, the matter will be escalated to the course coordinator before the next deliverable deadline.

Part 2: Problem-Space Brief

2.1 The Interaction Problem

This project investigates the usability of the FAST-NU library portal — the Insignia ILS system accessible at *nu.insigniails.com/library* — with a specific focus on two core student tasks: study room booking and book search. The portal is the primary and only self-service interface available to FAST-NU students for accessing library resources, yet it presents significant interaction barriers that cause students to fail tasks, make errors, or abandon the system entirely.

2.2 Why This Problem Is Worth Investigating

The Insignia library portal is a daily-use system with a captive user base — FAST-NU students have no alternative interface for booking study rooms or managing their borrowing. Despite this, the system suffers from two well-defined and observable usability failures:

Room Booking: The room booking feature is buried under a multi-level navigation path (My Account → Booking → Book Room) with no shortcut, prominent link, or home-page entry point. Once a student locates the feature, the booking process itself is difficult to complete: the date selection and the time-duration slider are unintuitive, error-prone, and not clearly labeled, resulting in incorrect bookings or task abandonment.

Book Search: The system presents four parallel search entry points — Simple, Advanced, Kiosk, and Assets — with no explanation of what each does or when to use which. Students attempting to find a specific book must navigate between these modes without guidance, and may not discover that the Advanced search supports filtering by author, subject, or call number. This creates unnecessary friction for a task that should be straightforward.

Both problems represent a genuine mismatch between what the system offers and how students can effectively use it. Because this is a real, actively used university system, improvements have direct consequences for students' academic work and time.

2.3 Early Assumptions (To Be Tested in Empathize)

The following are explicitly labeled as assumptions — things the team currently believes but has not yet verified with real user evidence. These will form the basis of our D1 research questions.

Assumption 1 (Room Booking): We assume that students find the study room booking process frustrating at multiple levels — both in navigating to the feature (buried under My Account → Booking → Book Room) and in completing the booking itself, particularly the date picker and duration slider which we believe are unintuitive and cause incorrect bookings. We have not yet confirmed whether the primary drop-off point is finding the feature or completing it, or whether students bypass the portal entirely and book rooms in person at the library desk.

Assumption 2 (Book Search): We assume students find the book search confusing because the system presents four parallel search modes — Simple, Advanced, Kiosk, and Assets — with no explanation of what each does or when to use which. We assume most students default to Simple search but struggle to filter or narrow results, and may not be aware that Advanced search supports filtering by author, subject, or call number. This needs to be confirmed — it is possible that students bypass the portal search entirely and ask library staff directly, or use Google to identify a book title before coming to the portal.

Part 3: Preliminary Stakeholder Map

This map represents the team's current understanding of who is affected by the library portal's interaction problems. It is expected to evolve once real fieldwork evidence is collected during the Empathize stage.

3.1 Primary Stakeholders

People who directly use the portal to accomplish tasks.

Stakeholder	Description
FAST-NU Undergraduate Students	The main users of the portal — search for books, check availability, manage loans and fines, and book study rooms. Directly affected by every usability failure in the system.
FAST-NU Postgraduate Students	Use the portal for research-related book searches and borrowing. May have different usage patterns and expectations compared to undergraduates.

3.2 Secondary Stakeholders

People who interact with the system occasionally or indirectly, or whose work is affected by how students use it.

Stakeholder	Description
Library Staff and Front-Desk Personnel	Manage reservations, handle student queries when the portal fails them, process returns and fines, and maintain room booking schedules. When the portal is hard to use, students fall back to asking staff directly — increasing their workload.
Library Administrators	Responsible for configuring the Insignia system, managing inventory, and approving room availability. Their backend decisions shape what students see on the front end.

3.3 Tertiary Stakeholders

People affected by the system without using it directly.

Stakeholder	Description
Faculty Members	Assign specific textbooks and expect students to locate and borrow them through the library. Do not use the portal themselves but are affected when students cannot access required materials efficiently.
University IT Department / Administration	Responsible for licensing and maintaining the Insignia ILS system. Decisions about the portal's design and configuration are made at this level. Any redesign recommendations from this project would need to be actionable within their constraints.
Future Students	Incoming students who will inherit the system as-is. Their library experience is shaped by a portal they had no input in designing.

Note: This stakeholder map is a starting hypothesis. Primary stakeholders may be refined after Empathize-stage fieldwork, and secondary/tertiary categories may expand once we understand the full scope of the booking and search problems.